Business Analytics ØABED2200

Exam Case 2026

Strategic Wealth Consulting

Group number: G06



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INTRODUCTION

Strategic Wealth Consulting, an Oslo-based advisory firm founded in 2006, was approached by a 35-year-old client, S. Nilsen, seeking guidance on how to invest an inheritance of NOK 5,000,000 received from her grandmother. The client is financially stable: she owns her apartment in Oslo, has a car, and maintains a secure job that fully covers her household expenses and the needs of her two children. Her objective is to invest the inherited capital over a 20-year horizon in order to optimise its value and provide long-term financial support for her children.

The core business problem therefore consists of identifying the most profitable investment strategy that balances expected return and risk, in line with the client's financial situation and risk preferences. To address this question, we evaluated three alternative investment options: (1) placing the funds in savings accounts, (2) purchasing a rental apartment, and (3) investing in the stock market. Using an Excel-based financial model, we analyzed official historical data and incorporated uncertainty through Monte Carlo simulation. In addition, we applied decision theory to interpret the simulated results, compare the alternatives under risk and uncertainty, and support a structured, data-driven recommendation. The Excel model is structured around a navigation menu page that provides direct hyperlink access to all underlying analysis sheets - covering interest rate history, real estate data, stock fund analysis, simulation outputs, and decision theory - allowing the reader to easily explore each component of the analysis.

Our results indicate that the different asset classes do not present equivalent risk-return profiles, as the real estate option consistently outperforms the alternatives across all simulated scenarios. This conclusion is further reinforced by the application of multiple decision-making criteria, including the Maximin and Maximax approaches, minimax regret, the Hurwicz criterion, the Laplace criterion, as well as expected monetary value (EMV) and the expected value of perfect information (EVPI). All these methods unanimously identify real estate as the most favorable investment option. The final recommendation is therefore to prioritize this option, as it maximizes expected returns while maintaining an acceptable level of risk and aligns closely with the client's financial objectives.

BACKGROUND

When individuals receive a large amount of capital, deciding how to invest it becomes an important financial decision. Different investment opportunities offer different levels of risk and return, and selecting the right strategy depends on the investor's goals, time horizon, and financial situation. Common investment vehicles include savings accounts, stocks, and real estate, among a wide array of other traditional and alternative assets.

We charge a one-time consultation fee of NOK 25,000, which the client covers from her own funds independently of the inherited capital, ensuring that the full NOK 5,000,000 remains available for investment. In addition, an ongoing annual management fee of 2.5% applies to both the real estate and stock investment options, reflecting the active advisory services provided by the firm throughout the 20-year horizon. For real estate, this covers tenant coordination, lease management, and annual performance reviews; for the stock portfolio, it encompasses continuous monitoring, periodic rebalancing, and investment advisory support. No management fee applies to the bank deposit strategy, as this option is self-managed once the funds are allocated across the four institutions. All reported investment returns are presented net of these fees, ensuring a fair and transparent comparison across the three alternatives.

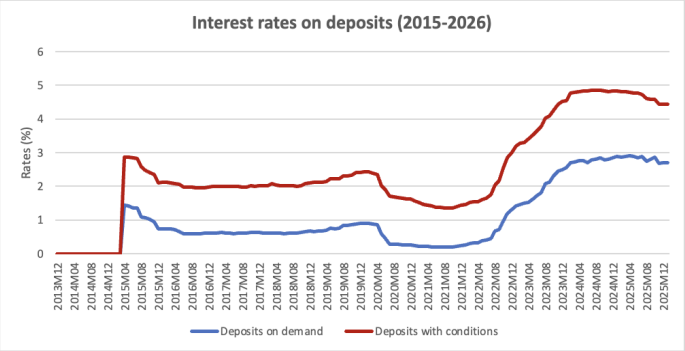
Taxation is also an important factor when comparing investment options. In Norway, several forms of investment income are subject to taxes. Rental income from property, interest earned from savings accounts, and capital gains from selling property are generally taxed as ordinary income. According to

the Norwegian Tax Administration (Skatteetaten, 2026a), the ordinary income tax rate is 22%, and this rate is therefore included in the model. For stocks, however, dividends and capital gains are taxed at a higher rate of 37,64%, and this rate is applied accordingly in the stock investment calculations. By incorporating taxes into the analysis, the comparison between the different investment alternatives becomes more realistic, as taxes directly reduce the final net return that the investor receives.

Banks

In the current low-to-moderate interest rate environment, households face a wide range of savings and investment options, each characterized by different levels of risk, liquidity, and expected return. Among the most common and accessible instruments in Norway are bank deposits, which can broadly be divided into deposits with variable interest rates (such as deposits on demand) and fixed-rate deposits (such as deposits with conditions).

Variable-rate accounts offer greater flexibility and liquidity, allowing depositors to withdraw funds at any time. However, this flexibility comes at the cost of uncertainty, as interest rates can fluctuate over time depending on market conditions. In contrast, fixed-rate deposits - commonly referred to as Fastrenteinnskudd - provide a predetermined interest rate over a specified investment horizon, offering greater predictability and typically higher returns. Data from Statistics Norway (2026a) confirms that fixed-rate deposits consistently offer more attractive interest rates compared to variable-rate alternatives over comparable periods. Moreover, given our objective of constructing an investment strategy with the lowest possible level of risk, fixed-rate products appear particularly suitable. This choice is further justified by the client's profile, characterized by a moderate risk tolerance and a long-term investment horizon of 20 years. In such a context, stability and predictability of returns are key factors.



Given the objective of minimizing risk, particular attention was paid to capital protection mechanisms. In Norway, the Norwegian Banks' Guarantee Fund guarantees deposits up to NOK 2 million per depositor per bank. To fully benefit from this protection, the investment was evenly distributed across four different banks. This diversification strategy ensures that the entire initial capital is covered by the guarantee scheme, while also protecting the accumulated interest over time. Assuming favorable interest rates, this approach makes it possible for the guaranteed amount to grow significantly, potentially reaching up to NOK 8,000,000 over a 20-year horizon while remaining fully protected. Additionally, Norwegian banks generally do not charge account opening or maintenance fees for standard savings accounts, meaning that allocating funds across multiple institutions does not generate additional costs.

Property

In Norway, real estate has traditionally been considered an attractive long-term investment. Housing demand has remained strong for many years, especially in larger cities where population growth and limited housing supply have contributed to rising property prices. Because of this, many investors

purchase apartments not only to generate rental income but also to benefit from the potential increase in property value over time. Real estate investments, therefore, often provide two types of financial benefits: regular rental income and capital gains when the property is sold.

In this project, we focus on a strategy where an apartment is purchased, rented out, and then sold after 20 years. This approach allows the investment to generate income during the ownership period while also capturing potential price growth in the housing market. By combining these two sources of return, the strategy aims to maximize the overall profit from the investment.

We also decided to focus on long-term rent rather than short-term rentals such as Airbnb. Long-term rent provides a more stable and predictable income stream because tenants typically sign contracts for longer periods and rental payments remain consistent. In contrast, short-term rentals depend heavily on tourism and seasonal demand, which can lead to fluctuating income and require significantly more management from the property owner.

Stocks

When saving for a longer duration such as 20 years, it is important to make strategic moves. Investing money in the stock market is deemed to be a promising option to maximize investments. Investing is generally regarded as more effective than traditional savings, as it enables capital to grow at a faster rate over time. This is due, among other things, to compound interest. Investing in funds also allows for higher returns and better risk diversification. Unlike putting your money in the bank, investing has its risks, however, it also offers the potential for exponential growth. In the long run, investing can be a valuable tool for optimizing your money (Surmount, 2024).

As the market will be for the most part unpredictable it is important to diversify the money so that we reduce the risk and will balance out any negative effects on our investment. Although investing carries its risk, it may be an efficient way to achieve financial goals over time, as long as you make informed decisions and have a strategic approach. Therefore, we in *Strategic Wealth Consulting* have suggested investing in Exchange – Trust Funds (ETFs). This way, any unpredicted plunges in a stock will be balanced out by the rest of the remaining stocks in the fund. ETFs and traditional funds are both types of investment funds that gather capital from multiple investors and allocate it in a diversified portfolio of securities. The ETF differs from other funds as it is bought and sold on an exchange throughout the day, much like stocks. Whereas traditional funds are typically purchased directly from a fund manager and priced once a day.

Given the client's moderate risk tolerance, we recommend a portfolio with equal allocations to two ETF funds, QQQ and VIG, representing different investment strategies within the equity market.

- Vanguard Dividend Appreciation Index Fund ETF Shares (VIG) is a low-risk exchange-traded fund (ETF) managed by Vanguard and consists of U.S. companies with a history of increasing their dividends over time. These firms are typically more mature and financially stable, offering lower volatility and more predictable returns compared to growth-focused funds (Yahoo Finance, 2026). A beta of 0.84 indicates that the fund is less volatile than the overall market. This is further supported by its relatively low standard deviation of 13.56%, suggesting lower variability in returns and therefore lower risk.
- Invesco QQQ Trust (QQQ) is an exchange-traded fund (ETF) that tracks the Nasdaq-100 Index, which consists of the 100 largest non-financial companies listed on the Nasdaq stock exchange. Nasdaq is a U.S.-based exchange with a strong concentration in the information technology sector, including major firms such as Meta and Microsoft. As a result, QQQ provides exposure to high-return opportunities but is also associated with higher volatility and risk (Yahoo Finance, 2026). A beta of 1.18 indicates that the fund is more volatile than the overall market, meaning

its returns tend to fluctuate more strongly. This is further supported by its higher standard deviation of 19.32%, reflecting increased variability in returns and a higher level of risk.

By combining VIG and QQQ, we achieve diversification across both sectors and investment styles. Diversification is a key principle in portfolio construction, as it reduces the risk of holding assets that are highly positively correlated. If investments respond similarly to negative market events, the overall portfolio risk increases significantly. By combining funds with different characteristics, we aim to achieve a more balanced and stable risk-return profile. In addition, the U.S. market is one of the most developed and innovation-driven economies in the world, making it an attractive component of a long-term investment strategy.

METHODS

Simulation

When analyzing investment decisions, uncertainty is unavoidable. Many factors that influence financial results cannot be predicted exactly, especially over a long time horizon. In our case, variables such as stock market returns, interest rates, rental income, or changes in property values may fluctuate over time. Because of this uncertainty, it would not be realistic to base the analysis on a single fixed value for each variable. For that reason, simulation is a suitable method to evaluate how the investment could develop under different future conditions (Nygaard, 2026b).

Simulation is particularly useful when a model includes uncertain input variables that influence a final outcome. Rather than assuming one expected value for each uncertain factor, simulation generates many possible scenarios based on probability distributions, allowing us to see how the investment might perform under different economic conditions. One of the main advantages of this approach is that it provides a more realistic picture of uncertainty compared to simpler methods — if we only look at average values or a few selected scenarios, important variations in the results could be missed. By automatically generating hundreds or thousands of possible outcomes, simulation makes it possible to observe the full distribution of results and better evaluate the risk associated with each investment option (Nygaard, 2026b).

To run the simulation, random number generators are used in the Excel model. These functions generate random values that follow specific probability distributions. A common example is the RAND() function, which creates random numbers between 0 and 1. These values can then be transformed into other distributions depending on how the variable behaves in reality. For example, if a variable is assumed to follow a normal distribution, the function NORM.INV() can be combined with RAND() to generate values around a given mean and standard deviation (Nygaard, 2026b). To represent uncertainty, variables can be described using probability distributions, where each possible outcome is associated with a specific probability. This allows the calculation of expected values and provides the basis for further quantitative analysis and simulations (Render et al., 2024, p. 78-79).

The simulation follows four main steps: identifying uncertain variables, assigning probability distributions and generating random values, replicating the model many times to create a large number of scenarios, and analyzing the results using descriptive statistics (Nygaard, 2026b).

In this project, simulation is used to evaluate the three investment alternatives available to the client: placing the money in savings accounts, investing in the stock market, and purchasing a rental apartment. For each option, the simulation generates a large number of possible outcomes over the 20-year investment period. This helps us compare the risk and return of the different strategies. In addition, simulation allows us to analyze a diversified portfolio strategy, where the client spreads her

investment across several asset classes, and to evaluate how diversification affects the distribution of long-term returns.

Decision Theory

While simulation helps us understand the range of possible outcomes for each investment option, it does not automatically tell us which alternative should be chosen. To interpret the results and compare the different strategies in a structured way, we apply decision theory.

Decision theory is a systematic approach to decision making that helps evaluate different alternatives based on data and logical reasoning. The main idea is to structure the decision process so that the alternatives, possible outcomes, and potential payoffs can be compared in a transparent way. Instead of relying purely on intuition, decision theory provides tools that support rational and well-informed decisions (Nygaard, 2026a).

The process usually starts by clearly defining the decision problem. In this case, the goal is to identify the investment strategy that best supports the client's objective of increasing the value of her inheritance over the next twenty years while keeping the level of risk acceptable. The next step is to identify the available alternatives. In our analysis, these include investing the inheritance in savings accounts, the stock market, or a rental apartment. After identifying the alternatives, it is necessary to consider the possible outcomes, often referred to as states of nature. These outcomes represent different economic conditions that may occur in the future and influence the performance of the investments. For example, financial markets may perform strongly or weakly, and property values may increase at different rates. Each alternative therefore leads to different financial results depending on these external conditions (Nygaard, 2026a).

Decision theory offers several criteria that help evaluate alternatives when future outcomes are uncertain. Decision theory under uncertainty applies when the decision maker does not know the probabilities of the various outcomes and must therefore choose based on the possible scenarios alone. The Maximax criterion focuses on the most optimistic scenario and selects the alternative with the highest possible payoff, assuming that the best possible outcome will occur. In contrast, the Maximin criterion focuses on the worst-case scenario and selects the option with the best minimum outcome, which reflects a more risk-averse perspective (Nygaard, 2026a). Another approach is the Hurwicz criterion, which represents a compromise between optimism and pessimism, using a coefficient to assign weights to the best and worst outcomes of each alternative. The Laplace criterion assumes that all possible outcomes are equally likely and therefore compares the average payoff of each option. Finally, the Minimax regret criterion focuses on minimizing potential regret, meaning the opportunity loss that could occur if the wrong decision is made (Nygaard, 2026a).

In addition to these criteria, decision theory under risk is applied when the decision maker knows the probabilities of the various outcomes. In this case, three methods are used: Expected Monetary Value (EMV), which calculates the probability-weighted average payoff for each alternative and selects the one with the highest expected return; Sensitility Analysis, which examines how changes in key assumptions or probabilities affect the decision outcome; and the Expected Value of Perfect Information (EVPI), which measures the maximum amount a decision maker would be willing to pay for complete and accurate information about future outcomes before making a decision (Nygaard, 2026a).

By combining simulation with decision theory, we are able to analyze the possible outcomes of each investment strategy and then evaluate them in a structured way. This approach helps us compare the alternatives more clearly and ensures that the final recommendation is based on quantitative analysis while also taking into account the client's moderate risk tolerance and preference for diversification (Nygaard, 2026a).

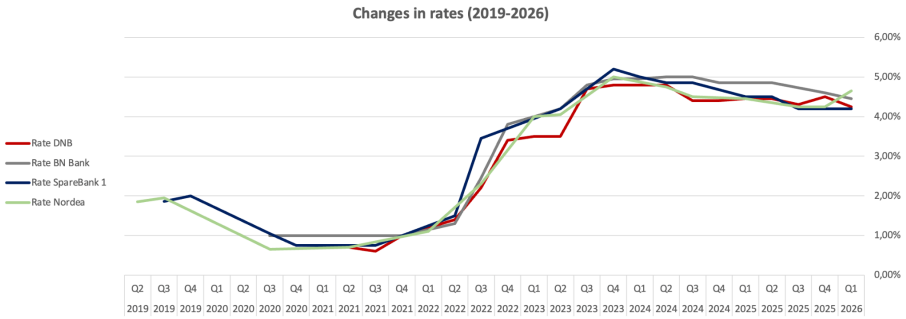
DATA

Banks

The analysis relies on the collection and construction of historical interest rate data for fixed-rate savings accounts offered by major Norwegian banks. One of the main challenges encountered was the limited availability of historical data, as banks generally only publish current interest rates and do not provide accessible archives of past rates. To overcome this limitation, historical data was retrieved using Wayback Machine, which allows access to previous versions of bank websites (Internet Archive, 2026). This method made it possible to collect past fixed-rate deposit offers and reconstruct a dataset of interest rates over time for each bank.

Several banks were initially considered, including DNB Bank ASA, Nordea Bank Norge, SpareBank 1, BN Bank ASA, Sbanken ASA, Santander Consumer Bank Norge, etc. However, only four banks—DNB, BN Bank, SpareBank 1, and Nordea—provided sufficient and consistent historical observations to allow for a reliable analysis of interest rate dynamics (DNB, 2026; BN Bank, 2026; SpareBank 1, 2026; Nordea, 2026). For most institutions, the available data primarily covers the period from 2020 onwards, with some additional observations from 2019. The frequency of archived data points is irregular, as the website snapshots are not recorded systematically (e.g., not every month). To ensure consistency while preserving as much information as possible, a quarterly frequency was adopted. In most cases, at least one observation per quarter was available, making this approach a reasonable compromise between data availability and precision. This method is also preferable to using annual averages, which would mask intra-year variations.

Descriptive statistics (minimum, maximum, mean, and standard deviation) were computed for each bank using Excel, and graphical representations were created to visualize the evolution and dispersion of interest rates over time. These statistics provided a basis for defining the range of possible interest rates for each institution. Given the absence of a clear theoretical distribution and the relatively limited dataset, no parametric distribution (such as the normal distribution) was imposed to simulate the evolution of the investment over a 20-year horizon. Instead, interest rates for each bank were assumed to vary within the empirically observed interval defined by their historical minimum and maximum values. This choice is economically justified. While recent years (post-2021) have been characterized by increasing interest rates - typically ranging between 3% and 5% - earlier periods, particularly during the COVID-19 crisis (2019–2020), exhibited significantly lower rates, often between 0% and 1%, and rarely exceeding 2%. Considering the 20-year investment horizon, it is essential to account for the possibility of future economic downturns that could lead to similar decreases in interest rates. At the same time, assuming a continuous increase leading to excessively high rates (e.g., above 10%) would not be realistic given long-term macroeconomic constraints and central bank policies. Finally, a uniform distribution bounded by the observed minimum and maximum values was implicitly adopted to model this uncertainty. This approach reflects a neutral assumption, where all values within the historical range are considered equally likely, while ensuring that simulated rates remain economically plausible and non-negative.



Property

Data for the model are collected from reliable Norwegian sources, such as Statistics Norway and Skatteetaten, covering housing prices, rental levels, taxation and interest rates, to reflect current market conditions. The purchase price is limited to NOK 4,700,000 based on observed listings from Finn.no (n.d.b), to allow for transaction costs and a remaining liquidity buffer within the initial capital of NOK 5,000,000. The investment is fully equity financed using the inherited NOK 5,000,000 to isolate the performance of the investment and not interfere with the rest of her finances. The property is modelled in St. Hanshaugen and Grünerløkka, areas characterised by high rental demand and relatively stable property values. These districts also have among the highest rental prices in Norway, supported by strong demand and limited housing supply, which is expected to support rental growth (NTB Kommunikasjon, 2025; Statistics Norway, 2026b; Husbanken, 2024). Stochastic variables are simulated using a normal distribution implemented in Excel as NORM.INV(RAND(), mean, SD), unless otherwise stated (Microsoft, n.d.).

In addition to the purchase price of NOK 4,700,000, transaction costs include a document tax of 2.5% (NOK 117,500) and a land registration fee of NOK 545 (Skatteetaten, 2026b; Kartverket, 2026). An additional NOK 100,000 is included for basic preparation, resulting in a total acquisition cost of NOK 4,918,045, and a remaining buffer of NOK 81,955. Property value development is modelled using the housing price index for Oslo and Bærum from Statistics Norway (2026b). Based on historical data, a mean and standard deviation are estimated and used to simulate annual price growth using a normal distribution. Property value evolves each year as the previous value is multiplied by (1 + the simulated growth rate) and is used to calculate wealth taxes and the final sales price.

Rental income is modelled using both historical and current market data. Historical rental statistics for Oslo and Bærum from Statistics Norway are used to simulate rental growth using a normal distribution (Statistics Norway, 2025a). As these data underestimate current market rent levels, they are only used for growth, while the initial rent level is based on current listings from Finn.no (n.d.a). Square meter price data for St. Hanshaugen and Grünerløkka are used to simulate apartment size using a normal distribution (Oslo kommune, 2024). The simulated apartment size compared to the purchase price is validated against comparable property listings from Finn.no (n.d.b). Based on this, comparable rental listings from Finn.no (n.d.a) are selected, and rent per square meter is calculated for each listing. Rental price per square meter is then simulated using a normal distribution based on these observations. Total rental income for all years is then calculated as the simulated apartment size is multiplied by the simulated rent per square meter and adjusted for the rental growth.

A vacancy interval of 0 to 14 days between tenants is assumed and simulated using a normal distribution, with MIN and MAX applied to prevent negative and unrealistic vacancy days. The chosen interval for vacancy was derived from general market knowledge, given the absence of direct empirical data for this specific variable. A notice period of three months is assumed, giving landlords time to find new tenants. Together with the limited supply and strong demand in districts such as St. Hanshaugen,

where apartments are typically re-rented quickly, this supports the expectation of low vacancy periods (Aftenposten, 2024; NTB Kommunikasjon, 2025; Husbanken, 2024).

Annual operating costs covering common expenses, insurance, and maintenance were included and modelled as a percentage of the property value, based on historical data. The starting cost for year one is simulated using a normal distribution, with MIN and MAX applied to keep values within the observed range of the historical percentages, ensuring realistic cost levels over time (Statistics Norway, 2025b).

Net rental income at the end of each year is placed in a savings account and earns interest over time. Based on the analysis of interest rates across the four banks presented in the previous section, BN Bank was selected, as the only other alternative with a higher maximum rate has a larger reduction in the minimum rate. The interest rate is simulated using the same approach as in the bank investment option, applying a uniform distribution between 1% and 5%, with tax on interest income deducted directly in the calculation to simplify the model while remaining realistic.

Based on the analysis of interest rates across four banks presented in the previous section, BN Bank was selected, as it offers a relatively favorable balance between minimum and maximum rates compared to the alternatives. The interest rate is simulated using the same approach as in the bank investment option, applying a uniform distribution between 1% and 5%. To simplify the model while maintaining a realistic representation, tax on interest income is deducted directly within the calculation, unlike other taxes in the model.

All other taxes are applied in separate columns, at the standard Norwegian income tax rate of 22%, covering rental income, capital gains from property sales, and interest income (Skatteetaten, 2026a, 2026d-f). Wealth tax is also included, as net wealth above NOK 1,900,001 is taxed with a municipal rate of 0.35% and a state rate of 0.65%, giving a combined rate of 1% (Skatteetaten, 2026c). This tax is included to reflect the tax burden associated with owning property and total wealth.

Renovation costs for the kitchen and bathroom were estimated using price ranges from Oppussingsguiden (2025a, 2025b), assuming standard and mid-level renovations. Costs are simulated using a normal distribution, with MIN and MAX applied to limit values within the defined range and adjusted for inflation. Inflation is captured in the historical growth rates for rental income and property values but is applied to renovation costs to reflect realistic price levels in the year they occur (Statistics Norway, 2026c). Renovation is assumed to take place in year 10. In practice, timing depends on tenant turnover and may occur earlier or later. As renovation may not align with the lease expiration, the impact is reflected through increased vacancy days, reducing rental income. The estimated duration is based on typical renovation timelines for Norwegian apartments, using bathroom renovation as it generally takes longer than kitchen renovation (Aktiv, 2023).

Selling costs at the end of the investment period are included and estimated between 1-3% of the property value based on guidance from DNB Eiendom (2025). These costs reflect typical expenses such as brokerage and administrative fees. Costs are simulated using a normal distribution, with MIN and MAX applied to keep values within the defined interval and are deducted from the final selling price to reflect transaction costs. The apartment is assumed to be sold at the end of year 20, with full rental income included for the final year. Additional vacancy at the point of sale is not modelled, as the effect is expected to be small relative to overall model uncertainty.

The model is based on historical data and therefore reflects past market conditions. While real data may deviate from normality, a normal distribution is used as a consistent and tractable approach to model uncertainty across variables. The approach ensures comparability across simulations and provides a structured way to capture variation around observed averages.

Stocks

To model the uncertainty associated with stock market returns, we constructed a simulation model based on realistic assumptions and historical patterns. The data used to estimate expected returns, and standard deviations were obtained from Yahoo Finance, which provides historical price data and key financial metrics for a wide range of financial assets.

This source is commonly used in financial analysis due to its accessibility and reliability. The analysis is based on five-year historical data, as this period provides a balanced perspective between short-term fluctuations and long-term trends. A five-year horizon allows the funds sufficient time to grow and reflect underlying performance, while still being recent enough to capture current market conditions. Shorter periods, such as one year, may be heavily influenced by temporary volatility or unusual market events, whereas longer periods, such as ten years, may include outdated conditions that are less relevant for current expectations.

Returns for both QQQ and VIG were modelled using probability distributions in Excel. Specifically, we used the NORM.INV function in combination with RAND() to generate random values that follow a normal distribution. This approach allows us to simulate realistic fluctuations in returns over time, centered around an expected average return with a given standard deviation. The use of a normal distribution is a common assumption in financial modelling, as returns tend to cluster around a mean value while still allowing for extreme outcomes. However, it is important to note that real-world financial data may deviate from perfect normality due to market shocks and other causes.

Dividend payments were also included in the simulation using predefined intervals in Excel. The dividend yields were estimated based on a five-year historical period. Since dividends are paid quarterly, the payments were summed to obtain annual yields, which were then averaged for each year. To define the intervals, the lowest and highest average annual yields observed over the five years were used as the lower and upper bounds. This approach reflects that dividend payments vary over time while ensuring that the simulated values remain within realistic limits. By using historically based intervals, the model captures both variation and stability in dividend behavior, contributing to a more accurate representation of total returns.

The parameters used in the simulation, such as expected return and standard deviation, were chosen based on historical performance and realistic market expectations. QQQ was assigned a higher expected return and higher standard deviation (1.34% and 19.26%) to reflect its growth-oriented and volatile nature. In contrast, VIG was assigned a lower expected return and standard deviation (1.05% and 13.56%), reflecting its focus on stable, dividend-paying companies.

Also, the limited liability feature of stock market investments was incorporated into the model by applying a non-negativity constraint on end-of-year capital, using the excel function max between 0 and the sum of capital start of year and return. This ensures that the portfolio value cannot fall below zero even if return is negative, reflecting the fact that investors cannot lose more than their initial investment.

RESULTS

To evaluate the investment options a Monte Carlo simulation model in excel is set up for each one to estimate how the investment value develops over a 20 year horizon. To ensure robust results and account for uncertainty in the variables, we performed 1000 simulation runs. This large number of repetitions allows us to capture a wide range of possible outcomes and construct a reliable distribution of return. Each run represents a possible scenario where key variables vary based on their underlying

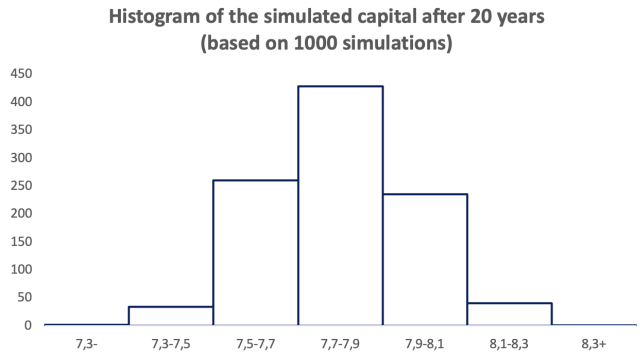
distribution. This produces a distribution of possible final investment values, allowing for an assessment of both expected outcomes and variability.

Banks simulation

The simulation results show a mean final value of NOK 7,791,263, representing the expected outcome across all scenarios. The median value of NOK 7,789,376 is nearly identical to the mean, indicating that the distribution of outcomes is approximately symmetric and that the typical result closely reflects the average. The standard deviation of NOK 168,793 is relatively low in relation to the mean, suggesting that outcomes are tightly concentrated and that the bank deposit strategy carries a limited degree of variability. The minimum simulated value is NOK 7,256,676 and the maximum is NOK 8,303,478, resulting in a range of approximately NOK 1,046,802. Even in the least favorable scenario, the investment generates a meaningful nominal gain over the initial capital, which reflects the low-risk nature of fixed-rate bank deposits.

The histogram confirms that the distribution of outcomes is narrow and approximately bell-shaped. Approximately 43% of all simulated outcomes fall within the NOK 7.7–7.9 million intervals, and nearly 96% of simulations produce a final value between NOK 7.5 million and NOK 8.1 million. The near-zero skewness (0.07) and slightly negative kurtosis (−0.13) further confirm that the distribution is symmetric with no pronounced tails, meaning that extreme outcomes in either direction are very unlikely.

Capital after 20 years (NOK)	
Mean	7 791 263,35
Standard Error	5 337,69
Median	7 789 375,72
Mode	#N/A
Standard Deviation	168 792,71
Sample Variance	28 490 979 897,17
Kurtosis	-0,13
Skewness	0,07
Range	1 046 802,29
Minimum	7 256 675,65
Maximum	8 303 477,95
Sum	7 791 263 354,48
Count	1 000,00
Confidence Level(95,0%)	10 474,38



A present value analysis was conducted to assess the real purchasing power of the projected nominal outcome under different inflation assumptions. Discounting the projected total capital after 20 years at a baseline inflation rate of 2% yields a present value of NOK 5,243,296, which suggests the investment maintains its real value above NOK 5,000,000. At a moderate inflation rate of 3%, the present value falls to NOK 4,313,833, reflecting a significant erosion of purchasing power. Furthermore, at a high inflation rate of 4%, the present value drops to NOK 3,555,830 - well below the initial capital - indicating a substantial real loss in purchasing power over the 20-year horizon.

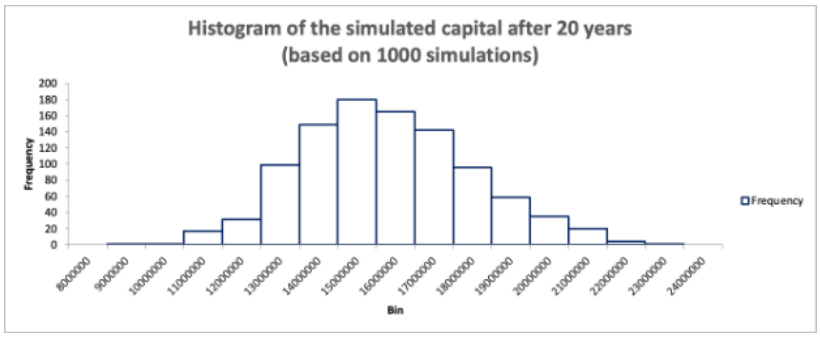
Overall, the results confirm that the bank deposit strategy offers a stable and highly predictable outcome in nominal terms with no extreme scenarios, consistent with its classification as the lowest-risk alternative. However, the present value analysis reveals that the strategy is sensitive to inflation. The results clearly illustrate that as inflation increases, the real value of the investment declines substantially, despite stable nominal growth. This highlights that capital preservation in nominal terms does not guarantee preservation in real terms. In other words, an investment with low variability and high predictability - such as bank deposits - may still lead to a loss of wealth in real terms if returns do not sufficiently outpace inflation.

Property simulation

The property model takes account for variables such as rental income, operating costs, renovation costs and taxes to calculate the final net value of the investment. The simulation results show a mean final value of NOK 15,276,053, representing the expected outcome across all scenarios. The median value is NOK 15,146,779, indicating that the typical outcome is close to the mean. The standard deviation of NOK 2,201,368 indicates variability in the outcomes. The minimum simulated value is NOK 8,404,736 and the maximum is NOK 22,357,500, resulting in a range of NOK 13,952,764. This shows that while most outcomes are concentrated around the mean, both significantly lower and higher outcomes are possible. A skewness of 0.27 indicates a slight positive skew, meaning that most outcomes are concentrated around the central range, with some scenarios resulting in particularly high values.

The histogram shows that approximately 65% of outcomes fall between NOK 13 million and NOK 17 million, with more than one third between NOK 14 million and NOK 16 million, representing the most likely range. Frequency decreases as values move further from this range, indicating that extreme outcomes are possible but less likely.

Capital after 20 years (NOK)	
Mean	15 276 053
Standard Error	69 613
Median	15 146 779
Mode	#I/T
Standard Deviation	2 201 368
Sample Variance	4 846 023 028 575
Kurstosis	-0,09
Skewness	0,27
Range	13 952 765
Minimum	8 404 736
Maximum	22 357 500
Sum	15 276 053 250
Count	1 000
Confidence Level (95,0%)	136 605



Overall, the results indicate that the property investment provides a stable outcome with a high expected value, although some variability remains due to uncertain market conditions. The distribution of outcomes forms the basis for further evaluation and comparison with alternative investment strategies using decision theory.

Stocks simulation

To evaluate the stock investment option, three strategies were simulated over a 20-year horizon: investing the full NOK 5,000,000 solely in VIG, investing the full NOK 5,000,000 solely in QQQ, and investing in a 50/50 portfolio with NOK 2,500,000 allocated to each fund. The portfolio approach was selected as the recommended strategy, as it combines the stability of VIG with the higher growth potential of QQQ, thereby achieving diversification across different investment styles.

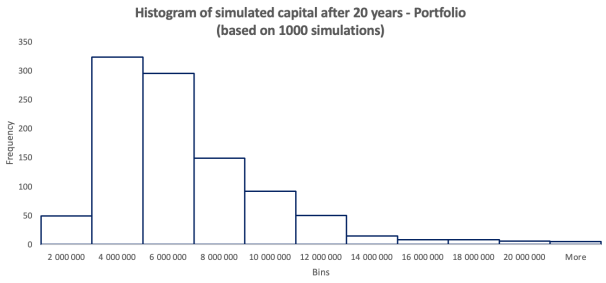
Focusing on the recommended 50/50 portfolio, the mean final capital is approximately NOK 5,581,195, indicating that the client can expect a substantial profit over the investment horizon. The median of approximately NOK 4,703,801 is somewhat below the mean, which is consistent with the right-skewed distribution produced by compounding returns over time. The standard deviation of approximately NOK 3,323,317 reflects the considerable variability inherent in equity investing: while the minimum observed outcome is approximately NOK 956,438, the maximum reaches approximately NOK 24,652,509. The wide range between minimum and maximum further highlights the high-risk, high-return nature of stock market investments.

Comparing the three strategies directly, VIG generates a higher mean and lower risk than QQQ when each is invested with the full NOK 5,000,000. The portfolio achieves the best outcome on both

dimensions simultaneously: the highest mean final capital (NOK 5,581,196) and the lowest standard deviation (NOK 3,323,318). This is the defining benefit of diversification in practice — combining assets that do not move in perfect lockstep reduces aggregate volatility and improves the compounded growth rate of the portfolio over time.

The distribution of portfolio outcomes is further illustrated by the histogram below, which shows the frequency of final capital values across all 1,000 simulation runs. The histogram of the simulated stock outcomes is not perfectly symmetric and therefore deviates from a normal distribution. Instead, it appears slightly right skewed, with a longer tail towards higher values. This can be explained by the nature of financial returns and the structure of the simulation. Although returns are generated using a normal distribution, the compounding effect over time leads to an uneven distribution of final capital. Positive returns accumulate and can result in very high values, while lower outcomes are more limited. As a result, the distribution becomes skewed rather than perfectly normal, which is consistent with real-world financial data.

	VIG	QQQ	Portfolio
Mean	5 307 555,48	5 144 798,71	5 581 195,99
Standard Error	109 375,21	168 041,04	105 092,53
Median	4 538 486,49	3 486 766,80	4 703 801,23
Mode	#I/T	#I/T	#I/T
Standard Deviation	3 458 747,97	5 313 924,25	3 323 317,60
Sample Variance	11 962 937 504 715,20	28 237 790 946 179,50	11 044 439 855 296,60
Kurtosis	8,61	16,60	4,74
Skewness	2,20	3,17	1,80
Range	28 679 558,94	57 740 567,86	23 696 071,30
Minimum	501 058,82	115 499,09	956 438,21
Maksimum	29 180 617,77	57 856 066,95	24 652 509,51
Sum	5 307 555 477,97	5 144 798 708,02	5 581 195 986,10
Count	1 000,00	1 000,00	1 000,00
Confidence Level (95,0%)	214 631,52	329 753,90	206 227,43



Decision Theory

To complement the simulation results and support a structured comparison of the three investment alternatives, decision theory criteria were applied to analyze the performance within the top 20% and bottom 20% of scenarios. To achieve this, the average of the best-case (top 20%) and worst-case (bottom 20%) outcomes from each simulation were calculated as the payoff values. This approach ensures that the decision analysis is grounded in simulated distributions rather than absolute extremes, making the results more representative of realistic outcomes. The three alternatives evaluated are bank deposits (NOK 8,026,916 best / NOK 7,556,756 worst), real estate (NOK 18,503,679 best / NOK 12,363,917 worst), and stocks (NOK 10,837,378 best / NOK 2,313,325 worst).

The Maximax criterion selects the alternative with the highest possible payoff, reflecting a fully optimistic decision maker who assumes the best outcome will occur. Under this criterion, real estate is the preferred alternative with a best-case value of NOK 18,503,679 - significantly higher than both stocks (NOK 10,837,378) and banks (NOK 8,026,916).

Maximax	Best scenario	Worst scenario	Max in a row
Banks	8 026 915,84	7 556 756,26	8 026 915,84
Real estate	18 503 678,89	12 363 917,41	**18 503 678,89**
Stocks	10 837 378,17	2 313 325,33	10 837 378,17

The Maximin criterion takes the opposite perspective, focusing on the worst-case scenario and selecting the alternative that performs best even under unfavorable conditions. Here, real estate again emerges as the preferred option, with a worst-case value of NOK 12,363,917. This is considerably higher than the worst-case outcomes for both banks (NOK 7,556,756) and stocks (NOK 2,313,325). Notably, the stock alternative carries the most severe downside risk, with a worst-case outcome barely above the initial investment of NOK 5,000,000 — reinforcing its high-risk profile relative to the other alternatives.

Maximin	Best scenario	Worst scenario	Min in a row
Banks	8 026 915,84	7 556 756,26	7 556 756,26
Real estate	18 503 678,89	12 363 917,41	**12 363 917,41**
Stocks	10 837 378,17	2 313 325,33	2 313 325,33

The Hurwicz criterion (Criterion of Realism) provides a balanced perspective by weighting the best and worst outcomes according to an optimism coefficient alpha. With alpha set at 0.8 - reflecting a moderately optimistic decision maker - the weighted averages are NOK 17,275,727 for real estate, NOK 9,132,568 for stocks, and NOK 7,932,884 for banks. Real estate again leads by a wide margin. This result holds across a wide range of alpha values, indicating that the dominance of real estate is robust and not sensitive to the specific level of optimism assumed.

Criterion of realism (Hurwicz, weighted average)	Best scenario	Worst scenario	Weighted average
Banks	8 026 915,84	7 556 756,26	7 932 883,92
Real estate	18 503 678,89	12 363 917,41	**17 275 726,59**
Stocks	10 837 378,17	2 313 325,33	9 132 567,60

The Equally Likely (Laplace) criterion assumes that all states of nature are equally probable and therefore selects the alternative with the highest average payoff across all scenarios. Real estate is again the preferred choice, and it is notable that banks outperform stocks under this criterion, driven by the stock alternative's relatively low worst-case outcome pulling down its average considerably.

Equally likely (Laplace)	Best scenario	Worst scenario	Average
Banks	8 026 915,84	7 556 756,26	7 791 836,05
Real estate	18 503 678,89	12 363 917,41	**15 433 798,15**
Stocks	10 837 378,17	2 313 325,33	6 575 351,75

The Minimax Regret criterion approaches the decision from the perspective of opportunity loss, selecting the alternative that minimises the maximum regret - that is, the potential loss from not having chosen the best alternative in each scenario. Real estate yields zero regret in both the best and worst states, as it is the dominant alternative in both cases. In contrast, banks carry a maximum regret of NOK 10,476,763 and stocks a maximum regret of NOK 10,050,592. The alternative with the minimum regret is therefore real estate, confirming that choosing any other option would result in substantial foregone value.

Minimax regret	Best scenario	Worst scenario	Max in a row
Banks	10 476 763,05	4 807 161,15	10 476 763,05
Real estate	0,00	0,00	**0,00**
Stocks	7 666 300,72	10 050 592,08	10 050 592,08

Across all five decision criteria under uncertainty - Maximax, Maximin, Hurwicz, Laplace, and Minimax Regret - real estate is consistently identified as the preferred investment alternative. This unanimity is particularly notable because these criteria represent fundamentally different attitudes toward risk and uncertainty, ranging from pure optimism to risk aversion and regret minimisation. The convergence of all criteria on the same alternative provides strong and robust support for the recommendation to prioritise real estate as the primary investment strategy for the client.

While the criteria under uncertainty assume no knowledge about the likelihood of different outcomes, decision making under risk incorporates explicit probability estimates for each state of nature. In this analysis, equal probabilities of 0.5 are assigned to the best-case and worst-case scenarios, reflecting a neutral assumption in the absence of reliable forecasts about future economic conditions. The Expected Monetary Value (EMV) criterion calculates the probability-weighted average payoff for each alternative and selects the one with the highest expected return. With equal probabilities of 0.5 assigned to both scenarios, the EMV for each alternative is equivalent to the simple average of its best and worst outcomes — which is consistent with the Laplace criterion applied earlier under uncertainty. The results

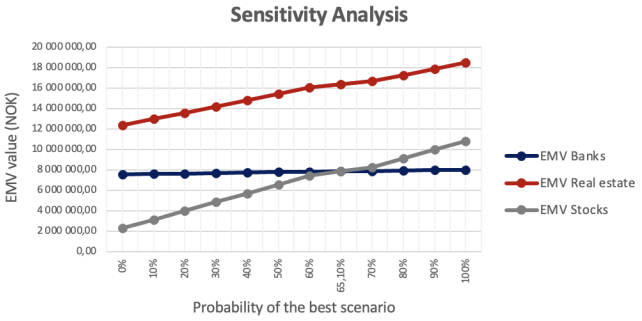
are as follows: real estate achieves the highest EMV of NOK 15,433,798, followed by banks at NOK 7,791,836, and stocks at NOK 6,575,352. Real estate therefore represents the superior choice under the EMV criterion, offering nearly twice the expected return of the bank deposit alternative and substantially outperforming the stock portfolio.

EMV	Best scenario	Worst scenario	EMV
Banks	8 026 915,84	7 556 756,26	7 791 836,05
Real estate	18 503 678,89	12 363 917,41	**15 433 798,15**
Stocks	10 837 378,17	2 313 325,33	6 575 351,75
Probabilities	50%	50%	100%

To examine the sensitivity of these results to the assumed probabilities, a scenario table was constructed showing how the EMV of each alternative changes as the probability of the best-case scenario varies from 0 to 100%. The analysis reveals that real estate dominates across all probability values. Even when the probability of the best-case outcome is set to zero — that is, assuming only the worst-case scenario occurs — real estate still yields the highest payoff at NOK 12,363,917. This finding further confirms the robustness of the real estate recommendation regardless of the probability assumptions made.

An additional insight from the sensitivity table concerns the indifference point between banks and stocks. The analysis shows that EMV(Banks) = EMV(Stocks) at a probability of approximately 65,10% for the best-case scenario, corresponding to an equal EMV of NOK 7,862,850 for both alternatives. Below this threshold, banks offer a higher EMV than stocks due to the stock alternative's severely low worst-case outcome. Above this threshold, stocks begin to outperform banks as the probability of the favourable scenario increases sufficiently to offset the downside risk. This crossover illustrates the asymmetric risk profile of stocks and highlights the importance of probability assumptions when comparing riskier investment options.

Probability	EMV Banks	EMV Real estate	EMV Stocks
0%	7 556 756,26	12 363 917,41	2 313 325,33
10%	7 603 772,22	12 977 893,56	3 165 730,61
20%	7 650 788,18	13 591 869,70	4 018 135,89
30%	7 697 804,14	14 205 845,85	4 870 541,18
40%	7 744 820,09	14 819 822,00	5 722 946,46
50%	7 791 836,05	15 433 798,15	6 575 351,75
60%	7 838 852,01	16 047 774,30	7 427 757,03
65,10%	**7 862 850,37**	16 361 166,22	**7 862 850,37**
70%	7 885 867,96	16 661 750,45	8 280 162,32
80%	7 932 883,92	17 275 726,59	9 132 567,60
90%	7 979 899,88	17 889 702,74	9 984 972,88
100%	8 026 915,84	18 503 678,89	10 837 378,17



The Expected Value of Perfect Information (EVPI) measures the maximum amount a decision maker should be willing to pay to obtain completely reliable information about which state of nature will occur. With perfect information, one would always choose the best alternative for whichever scenario materialises — selecting real estate in the best-case scenario (NOK 18,503,679) and real estate again in the worst-case scenario (NOK 12,363,917), since real estate dominates both states. The expected value with perfect information (EVwPI) is therefore NOK 15,433,798, which is exactly equal to the maximum EMV already achieved by real estate without any additional information. As a result, EVPI = EVwPI – Maximum EMV = NOK 15,433,798 – NOK 15,433,798 = NOK 0. This result confirms that perfect information has no additional value in this decision problem, because real estate is already the dominant alternative in both states of nature. No additional information could change the optimal decision, and therefore there is nothing to be gained from acquiring it.

EVPI	Best scenario	Worst scenario	EMV
Banks	8 026 915,84	7 556 756,26	7 791 836,05
Real estate	18 503 678,89	12 363 917,41	**15 433 798,15**
Stocks	10 837 378,17	2 313 325,33	6 575 351,75
Probabilities	50%	50%	100%
With perfect information	18 503 678,89	12 363 917,41	**15 433 798,15**

Together, the EMV analysis and the EVPI calculation reinforce the conclusions drawn from the criteria under uncertainty. Real estate is the optimal investment alternative not only under optimistic or pessimistic assumptions, but also when probabilities are explicitly incorporated into the analysis. The EVPI of zero further underscores the strength of this result: the recommendation to invest in real estate is unambiguous and would not be revised even with access to perfect foresight about future market conditions.

CONCLUSION

We evaluated three investment strategies for S. Nilsen's NOK 5,000,000 inheritance over a 20-year horizon: fixed-rate bank deposits, a rental apartment in Oslo, and a diversified stock portfolio (QQQ and VIG). The simulation results revealed that the three asset classes differ considerably in terms of both expected return and risk. Bank deposits yielded a stable mean of NOK 7,791,263 but are vulnerable to inflation changes. Stocks delivered a lower mean of ~NOK 5,581,196 with severe downside risk (worst-case ~NOK 956,438). Real estate consistently outperformed both, with a mean of NOK 15,276,053 and a worst-case of NOK 8,404,736 - well above the initial capital.

Decision theory confirms this unanimously. All five criteria under uncertainty (Maximax, Maximin, Hurwicz, Laplace, Minimax Regret) identify real estate as the preferred alternative. Under risk, real estate achieves the highest EMV of NOK 15,433,798, and the EVPI equals zero — meaning perfect information would not change the decision. These findings show that in the Norwegian market, real estate in Oslo uniquely combines high expected returns with contained downside risk, making it the clearly superior choice for a moderate-risk investor. Strategic Wealth Consulting therefore recommends that S. Nilsen invest her inheritance in a rental apartment in St. Hanshaugen or Grünerløkka, Oslo.

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